BIOCHEMISTRY CUMULATIVE EXAM

November 14, 1992

A liver cell suspension is prepared that is lacking both carbon and nitrogen metabolites. These liver cells are then supplied with glutamic acid as the sole source of both carbon and nitrogen.

A. (40) Give a complete enzymatic pathway for the synthesis of aspartic acid in these liver cells using glutamic acid as the sole source of carbon and nitrogen.

B. (30) Write a complete balanced equation for this conversion, including any ATP's generated, oxygen consumed, etc.

C. (30) If the glutamic acid was radiolabeled with C-14 at carbon number 3, which carbon of aspartic acid would first become labeled as a result of the above pathway.

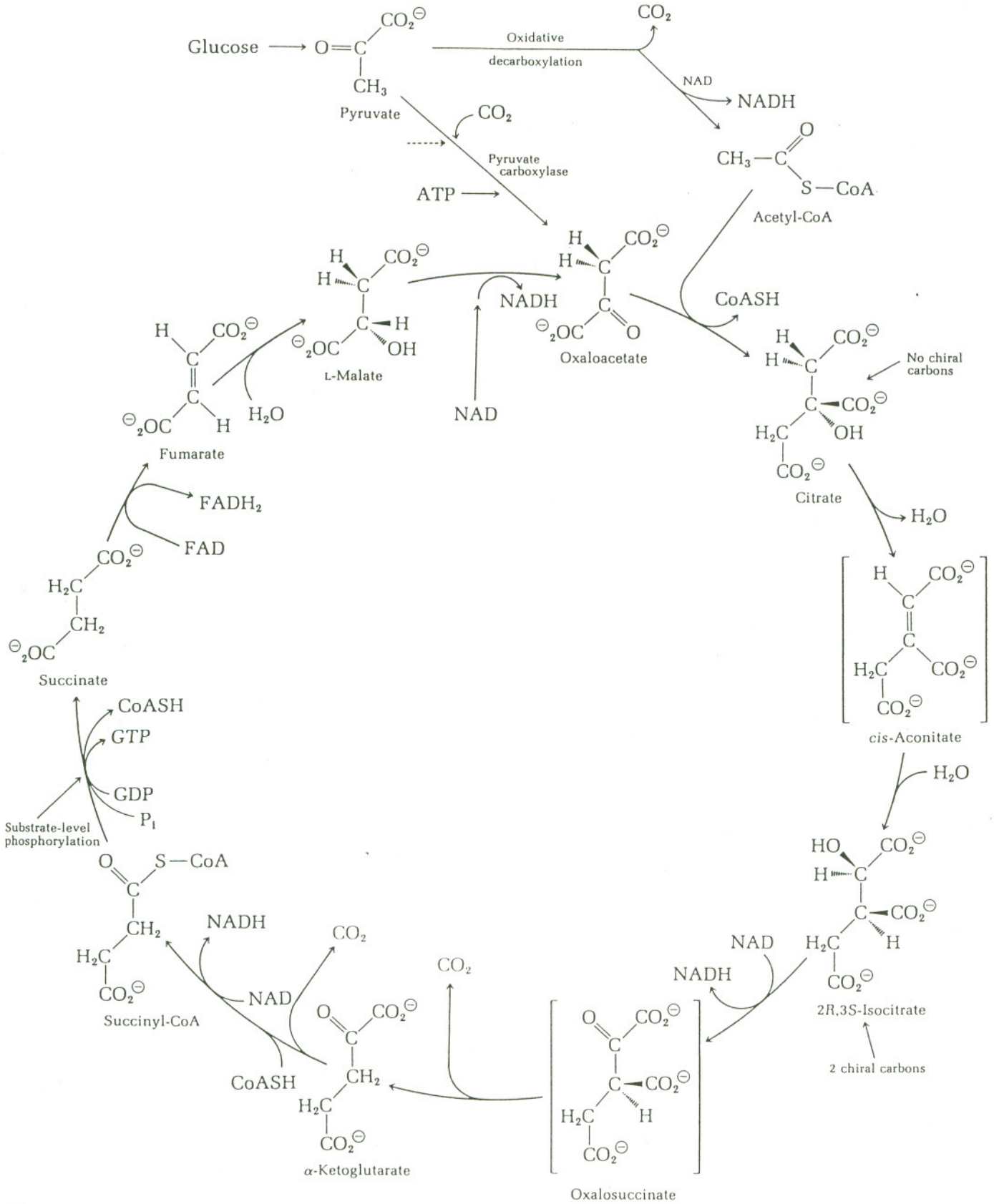


Figure 15.4 The citric acid cycle.

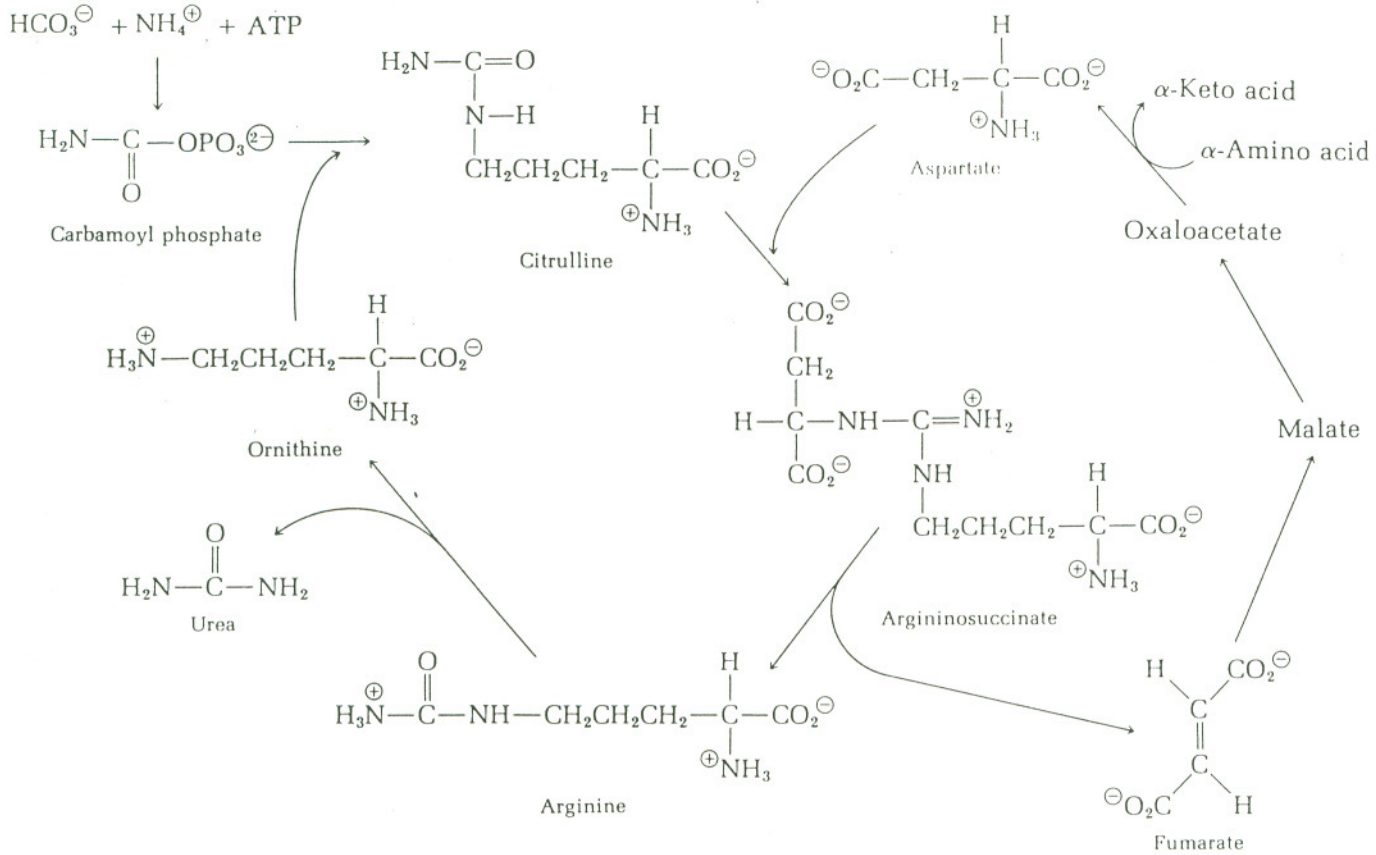
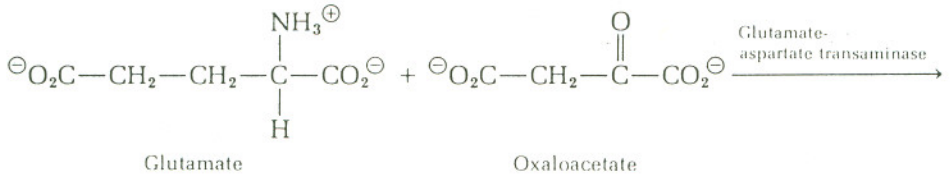
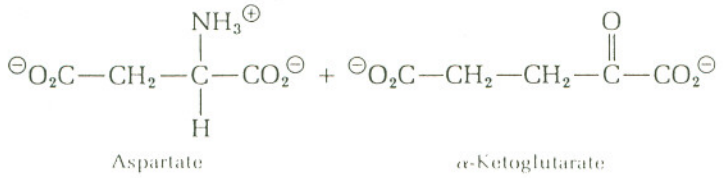
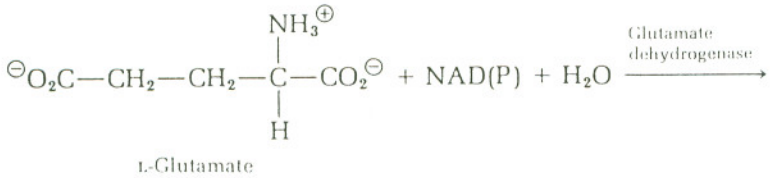


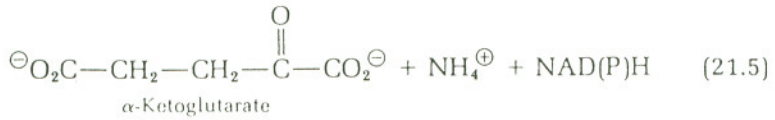
Figure 21.6 Interaction of the urea cycle with the citric acid cycle.





(21.4)





(21.5)